

DevOps Documentation

Program 9: Organize Kubernetes Pods Using Labels and Namespaces

Program-9

—> namespaces.yaml

—> pods.yaml

S1. Create a file for namespaces as namespaces.yaml

```
! namespaces.yaml x ! pods.yaml
! namespaces.yaml
1  apiVersion: v1
2  kind: Namespace
3  metadata:
4    name: devel
5  ---
6  apiVersion: v1
7  kind: Namespace
8  metadata:
9    name: produc
10
```

S2. Create a file for Pods as pods.yaml

```
! namespaces.yaml ! pods.yaml x
! pods.yaml
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: frontend-pod
5    namespace: devel
6    labels:
7      app: frontend
8  spec:
9    containers:
10   - name: frontend
11     image: nginx
12     ports:
13       - containerPort: 80
14   ---
15   apiVersion: v1
16   kind: Pod
17   metadata:
18     name: backend-pod
19     namespace: produc
20     labels:
21       app: backend
22   spec:
23     containers:
24       - name: backend
25         image: nginx
26         ports:
27           - containerPort: 80
```

S3. Apply the Namespaces & Pod YAML file

kubectl apply -f namespaces.yaml

kubectl apply -f pods.yaml

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program9$ kubectl apply -f namespaces.yaml
namespace/devel created
namespace/produc created
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program9$
```

S4. Verify Pods by Namespaces

kubectl get pods -n devel

kubectl get pods -n produc

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program9$ kubectl get pods -n devel
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           63s
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program9$ kubectl get pods -n produc
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           71s
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program9$
```

S5. Check the status

kubectl get pods

S6. Filter Pods by Label

⇒ List all frontend pods in dev namespace

kubectl get pods -n devel -l app=frontend

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program9$ kubectl get pods -n devel -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           2m37s
```

⇒ List all backend pods in prod namespace

kubectl get pods -n produc -l app=backend

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program9$ kubectl get pods -n produc -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           3m24s
```